

// Day 23 of 30

MOST DEVS GET THIS WRONG

SQL Query Execution Order

→ Swipe through all 6 slides



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How You Write SQL

This is NOT execution order

```
SELECT name, COUNT(*)  
FROM orders  
JOIN users ON orders.user_id = users.id  
WHERE status = 'active'  
GROUP BY name  
HAVING COUNT(*) > 5  
ORDER BY COUNT(*) DESC  
LIMIT 10
```

💡 SQL is written in one order, executed in another



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Actual Execution Order

How the database REALLY runs it

1. FROM → pick the tables
2. JOIN → combine tables
3. WHERE → filter rows
4. GROUP BY → group remaining rows
5. HAVING → filter groups
6. SELECT → pick columns
7. ORDER BY → sort results
8. LIMIT → cut to N rows

💡 FROM first. SELECT is almost last!



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Why This Matters

Common errors explained

Error: Can't use alias in WHERE

```
SELECT price * qty AS total  
FROM orders  
WHERE total > 100; -- FAILS!
```

Why? WHERE runs BEFORE SELECT
→ 'total' alias doesn't exist yet

Fix: WHERE price * qty > 100

💡 WHERE runs before SELECT — aliases don't exist yet



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WHERE vs HAVING

Now it makes sense

WHERE:

- Runs BEFORE GROUP BY
- Filters individual rows
- Can't use aggregate functions

HAVING:

- Runs AFTER GROUP BY
- Filters groups
- CAN use aggregate functions

```
WHERE status = 'active' -- filter rows
HAVING COUNT(*) > 5     -- filter groups
```

💡 WHERE = before grouping. HAVING = after grouping.



Performance Impact

Execution order affects speed

Bad: Filter late

```
FROM orders (10M rows)
JOIN users (expensive on 10M)
WHERE date > '2025-01-01'
```

Good: Filter early

```
FROM orders
WHERE date > '2025-01-01' (filters to 1M)
JOIN users (cheaper on 1M)
```

- WHERE reduces rows BEFORE join
- 10x faster

💡 Filter early, join late = fast queries



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Interview Questions

You'll get asked these

Q: Can you use alias in WHERE?

A: No. WHERE runs before SELECT.

Q: WHERE vs HAVING?

A: WHERE filters rows, HAVING filters groups.

Q: What runs first, JOIN or WHERE?

A: JOIN first, then WHERE.

Q: How to optimize slow queries?

A: Filter early (WHERE before JOIN)

💡 Know execution order = ace SQL interviews



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done = true

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